

EMMONAK, AK, USA

WMO: 702084

Lat: 62.785N

Lon: 164.491W

Elev: 14

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 96-19

WBAN: 26650

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.3	-18.5	-28.5	1.1	-16.8	-26.7	1.2	-14.9	36.2	17.8	32.8	23.2	3.2	10	0.856

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	68.4	58.6	64.4	56.2	62.7	55.3	59.6	65.8	57.8	63.6	56.0	61.3	9.4	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.7	68.6	61.7	54.7	63.8	60.2	53.7	61.3	59.1	26.1	66.2	24.9	63.2	23.7	61.0	68.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.5	25.3	22.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.6	8.0	12.0	11.9	22.1	37.7	50.4	53.5	52.2	45.2	32.0	19.9	9.2
	DBStd	19.91	15.41	14.14	12.98	10.64	8.42	5.80	4.53	4.77	5.36	7.26	10.90	14.34
	HDD50	7761	1302	1065	1182	837	389	62	15	27	155	559	904	1264
	HDD65	12927	1767	1485	1647	1287	846	439	357	398	594	1024	1354	1729
	CDD50	308	0	0	0	0	8	73	123	93	11	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	6	0	0	0	0	0	4	0	2	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	10.5	11.2	13.1	11.5	11.8	10.0	9.3	9.1	9.2	9.5	9.9	10.9	11.0
Precipitation	PrecAvg	24.4	1.7	1.7	1.3	1.1	1.1	1.5	2.7	3.7	3.1	2.6	2.2	1.8
	PrecMax	33.3	3.4	5.3	3.5	2.6	2.7	2.5	5.6	7.4	6.4	5.5	4.6	3.4
	PrecMin	17.6	0.5	0.4	0.3	0.3	0.4	0.7	0.7	1.1	0.8	1.2	0.4	0.7
	PrecStd	3.7	0.8	1.1	0.8	0.6	0.6	0.5	1.1	1.3	1.3	1.2	1.1	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	36.7	37.1	43.0	64.2	73.3	72.1	72.2	61.1	48.2	42.5	39.0
		MCWB	33.6	33.9	34.0	37.3	52.3	58.1	60.7	62.1	52.1	45.9	39.8	34.7
	2%	DB	35.8	35.0	35.7	39.0	59.3	68.0	67.8	66.3	56.9	45.4	37.4	36.1
		MCWB	33.3	33.2	33.4	35.2	49.2	56.9	58.7	58.7	50.8	42.3	35.4	33.8
	5%	DB	33.5	33.9	33.7	36.8	54.6	63.8	64.1	63.2	54.6	43.3	36.3	33.8
		MCWB	31.7	32.4	32.1	33.9	47.3	54.7	57.0	56.6	50.1	40.0	34.5	32.2
	10%	DB	28.4	31.8	30.4	35.8	50.8	61.0	61.4	59.2	52.4	41.1	34.2	28.4
		MCWB	27.2	30.8	29.5	33.4	45.3	53.2	55.8	54.5	48.9	38.4	32.7	27.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.5	34.7	34.8	38.3	53.3	60.1	61.7	64.2	53.5	46.5	40.1	36.1
		MCDB	37.4	35.7	36.3	41.7	62.6	68.9	68.9	69.2	57.6	47.4	42.2	38.1
	2%	WB	33.1	33.3	33.2	35.8	50.3	57.8	59.5	60.0	51.9	42.8	36.9	33.6
		MCDB	35.2	34.2	34.6	38.6	58.2	65.6	65.5	65.5	55.7	45.3	37.9	35.5
	5%	WB	31.5	32.4	31.9	34.3	47.9	55.6	58.0	57.5	50.6	40.9	34.2	31.8
		MCDB	33.1	33.6	33.3	36.8	54.7	63.0	63.5	62.2	54.0	43.8	35.5	33.6
	10%	WB	27.8	30.3	29.6	33.0	45.2	53.6	56.4	55.1	49.4	38.9	32.4	27.9
		MCDB	28.4	31.5	30.5	35.1	50.4	60.0	61.4	59.2	52.4	41.5	33.9	28.5
Mean Daily Temperature Range	5% DB	MDBR	11.6	12.6	13.3	12.7	11.6	12.0	9.6	9.2	9.1	8.1	9.3	11.3
		MCDBR	10.4	11.0	10.7	11.8	19.3	18.3	15.6	15.3	12.6	9.8	7.1	10.1
		MCWBR	9.3	10.0	9.3	9.5	11.6	9.6	8.8	9.0	8.0	7.5	6.5	9.0
	5% WB	MCDBR	10.5	10.6	10.0	11.0	18.5	17.3	14.6	14.4	10.8	8.8	7.3	10.1
		MCWBR	9.5	10.2	9.0	9.0	11.2	9.3	8.6	9.0	7.4	7.2	7.0	9.3
Clear-Sky Solar Irradiance	taub	0.205	0.273	0.280	0.349	0.372	0.366	0.370	0.396	0.328	0.292	0.227	0.196	
	taud	1.999	2.125	2.203	2.052	2.208	2.404	2.435	2.309	2.507	2.382	1.998	1.886	
	Ebn at Noon	143	210	262	260	266	272	267	246	247	208	123	71	
	Edh at Noon	10	21	29	42	39	33	31	33	22	16	10	6	
All-Sky Solar Radiation	RadAvg	78	300	854	1429	1661	1675	1328	1061	679	314	87	33	
	RadStd	11	38	95	101	172	172	71	133	60	28	15	3	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (10 neighbors)	+1.49	N/A	N/A	N/A	+0.86	+0.99	N/A	N/A	+63	N/A

Nomenclature: See separate page